Government Information Quarterly 39 (2022) 101745

Contents lists available at ScienceDirect

Government Information Quarterly

journal homepage: www.elsevier.com/locate/govinf

# Measuring the performances of politicians on social media and the correlation with major Latin American election results

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ARTICLE INFO

*Keywords:*
Social media
Elections
Social networks
Latin America
Democracy

ABSTRACT

Contemporary social media (SM) has strongly impacted democratic practices. The success of presidential campaigns is frequently attributed to being highly correlated with the candidates' social media performance, but there is no well-established method to measure this performance. Thus, this study aims to improve the understanding of a politician's performance on SM and its correlation with electoral results. Applying a new, recently-defined set of metrics, based on Zajonc's exposition theory and considering the interactions of users on politicians' profiles in multiple SM platforms, this research identifies statistical correlations between SM performance and the votes received in multiple elections. As case studies, this paper focuses on the most recent presidential elections in the four most populous countries in Latin America: Argentina (2019), Brazil (2018), Colombia (2018), and Mexico (2018). Data from more than 65,000 posts were collected from the SM profiles of the main candidates on Facebook, Twitter, and Instagram, starting from 300 days before the election days, and correlations with electoral results were calculated. The results demonstrated strong correlations between the defined metrics and the votes received, particularly the engagement per post, although there were differences among countries. On the other hand, we observed that there is zero or negative correlation between the number of posts and the electoral results.

## 1. Introduction

Social media (SM) has played a central role in politics and elections throughout the last decade. We have entered a new era mediated by SM in which politicians conduct permanent campaigns with neither geographic nor time constraints, and any extra information about them may be obtained not only through the press but directly from their profiles on social networks and through other people sharing and amplifying their voices on SM. In this new scenario, SM is used extensively in campaigns, and the success of campaigns is frequently attributed to being highly correlated with the success of candidates on SM, such as the campaigns of Obama (2008 and 2012) (Bimber, 2014; Cogburn & Espinoza-Vasquez, 2011) and Trump (2016) (Francia, 2018) in the U.S., the Brexit campaign in 2016 (Hall, Tinati, & Jennings, 2018), and the Bolsonaro campaign for the Brazilian presidency in 2018 (Brito, Paula, Fernandes, & Meira, 2019).

As a consequence, much academic research has been devoted to this modern political campaign paradigm and its activities (Carlisle & Patton, 2013; Jungherr, 2016), such as how well Facebook and Twitter users reflect the general voting public (Mellon & Prosser, 2017), or how the sentiment of conversations is connected to real-world events affecting a political campaign (Park, 2013), and whether it is possible to predict election results based on SM analysis (Tumasjan, Sprenger, Sandner, & Welpe, 2010).

The simple measuring of user engagement with SM remains a challenging task (Sigerson & Cheng, 2018). Also, although many initiatives have aimed at studying the SM behavior of candidates, the engagement of citizens on SM, and its correlation with electoral results, this area of research is still in its infancy and presents many challenges, both from psychological and technical viewpoints. As a result, very recent surveys (Chauhan, Sharma, & Sikka, 2021; Brito, Silva Filho, & Adeodato, 2021) have demonstrated that the most common approach, based on counting the volume of mentions on Twitter, followed by sentiment analysis, has frequently been criticized and contested. Thus, new studies based on novel approaches for measuring and correlating SM performance are needed.

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https://doi.org/10.1016/j.giq.2022.101745
Received 16 September 2021; Received in revised form 11 June 2022; Accepted 1 July 2022
Available online 25 July 2022
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### 1.1. The use of social media in elections

The first presidential campaign identified as being firmly based on SM platforms was the 2008 Barack Obama campaign in the U.S. (Bimber, 2014; Cogburn & Espinoza-Vasquez, 2011), through Facebook, Twitter, MySpace, e-mails, an iPhone application, and two websites. His campaign capability to translate online activity to on-the-ground activism (Cogburn & Espinoza-Vasquez, 2011) inspired the adoption of SM as a permanent campaign platform across the entire world.

After the Obama campaign, the success of many others has been attributed to their online campaigns, such as those for Brexit (Hall et al., 2018) and for Trump in the 2016 U.S. presidential elections (Francia, 2018; Hall et al., 2018). In the 2018 Brazilian presidential election, Bolsonaro had almost no TV time, but his SM profiles received more than half (55%) of all the interactions received by all candidates, and he was elected. On the other hand, the candidate with the longest TV time and fewer SM followers ended in fourth place (Brito, Paula, Fernandes, & Meira, 2019).

The impact of SM on politics and elections around the world has been receiving attention. Smyth (Smyth & Best, 2013) studied how SM was used in the 2011 elections in West Africa, Nigeria, and Liberia, concluding that SM helped to overcome a previous scarcity of information during the electoral process. Regarding the 2013 national election in Norway, Kalsnes (Kalsnes, 2016) described a "social media interaction deadlock," a phenomenon that is increasing the disparity between the expressed strategies and online performances of parties. In the 2014 Indian general elections, Jaidka (Jaidka & Ahmed, 2015) studied official Twitter accounts of the top ten political parties and identified the new paradigms created by political parties to engage and inform voters, driven by modern information and communications technology (ICT).

More recently, Hall (Hall et al., 2018) argued that discussions on SM only represent a small portion of the overall discussions in a political campaign and play a minor role in the overall ecosystem. However, in the same year, and regarding the 2016 U.S. presidential election, results by Morris (Morris, 2018) suggested that campaign messages about candidates sent *via* Twitter—regardless of the candidate of focus—resonated just as strongly with potential voters as those sent *via* traditional media, thereby reinforcing the power of SM. In one of the very few studies including Instagram, Aminolroaya (Aminolroaya & Katanforoush, 2017) highlighted that the flow of information on Instagram played a significant role in the Iranian parliamentary election.

With the increased use of SM in campaigns worldwide, researchers have naturally started to study the correlations of SM use and electoral performance, as presented below.

### 1.2. Social media and electoral performance

The correlation between SM performance and electoral performance has been the focus of research since 2010. Tumasjan et al. (Tumasjan et al., 2010) presented a seminal study in the German federal election aiming to predict the results based on the analysis of SM data. They collected all tweets containing the names of the six parties or selected prominent politicians represented in the German parliament and observed that the relative volumes of tweets closely mirrored the federal election results. They then claimed that "*the mere number of tweets mentioning a political party has high correlations with votes, can be considered a plausible reflection of the vote share, and has predictive power even comes close to traditional election polls*." In the same year, O'Connor (O'Connor, Balasubramanyan, Routledge, & Smith, 2010) reported similar results with a similar approach improved by sentiment detection of tweets.

With regard to the Dutch national elections of 2010, Kruikemeier (Kruikemeier, 2014) demonstrated that candidates who used Twitter during the campaign received more votes than those who did not, and that using Twitter in an interactive manner also had a positive impact. In the same elections, Effing (Effing, Van Hillegersberg, & Huibers, 2011) reported that politicians with higher SM engagement received relatively more votes within most political parties. In 2013, DiGrazia et al. (DiGrazia, McKelvey, Bollen, & Rojas, 2013) reported a statistically significant association between tweets that mention a candidate for the U.S. House of Representatives and his or her subsequent electoral performance. Later, Ramadhan (Ramadhan, Nurhadryani, & Hermadi, 2014) analyzed the use of SM in the 2014 Jakarta legislative election, indicating that the use of SM, especially Facebook and Twitter, was strongly correlated with the number of votes gained by the candidate. More recently and related to a Latin American election, (Olivares, Cárdenas, Losada, & Borondo, 2019) studied the correlation of the election outcome and the Twitter results for the second round of Chilean elections by analyzing a polarization metric.

Following these works, studies have been published aiming at correlating SM and election performance, ultimately trying to predict election results. Two very recent surveys have summarized these initiatives. Chauhan et al. (Chauhan et al., 2021), surveyed 38 papers, and Brito et al. (Brito, Silva Filho, & Adeodato, 2021) performed a systematic review of 83 relevant studies. Both reviews highlighted that most studies used Twitter as a corpus for correlating SM and election results based on volume and sentiment analysis. However, some studies challenge this approach (Gayo-Avello, Metaxas, & Mustafaraj, 2011; Jungherr, Jürgens, & Schoen, 2012; Jungherr, Schoen, Posegga, & Jürgens, 2017), and Brito et al. also observed that only 55% of the 64 studies that used this approach achieved success. Both survey studies (Chauhan et al., 2021; Brito, Silva Filho, & Adeodato, 2021) also highlight that the existence of irrelevant, junk, fake, or spam posts may affect the results, as well as the challenges of accurate sentiment analysis on tweets, which are small texts. Moreover, (Brito, Silva Filho, & Adeodato, 2021) highlighted the difficulties in using data from multiple networks besides Twitter, and the positive effect of collecting data for extended periods.

In order to deal with some of the challenges previously presented, (Brito, Meira, & Adeodato, 2021) defined a new model for measuring SM performance and applied it in the 2018 Brazilian presidential election. Their findings demonstrated a high correlation between the performance of politicians and electoral results. Although promising, their model was applied in only one electoral context, and its use in more scenarios is needed to better validate both the model and the conclusions.

### 1.3. The current study

Within this context, this research study aims to improve the understanding of the performance of politicians on SM and its correlation with electoral results. It aims to answer two main research questions:

**RQ1**. *Is there a correlation between the SM performance of candidates and their electoral performance?*

**RQ2**. *Is there a common pattern among countries between the SM performance of candidates and electoral results?*

These questions are answered taking into consideration the challenges presented in the surveys (Chauhan et al., 2021; Brito, Silva Filho, & Adeodato, 2021). Thus, we extended the model defined by (Brito, Meira, & Adeodato, 2021), based on how many people pay attention to a candidate through the number of interactions on the posts of the candidates' official profiles, and applied it to different elections, analyzing and comparing the results on the four most populous Latin American countries.

## 2. Materials and methods

To answer RQ1 and RQ2, this study considered a postpositivist stance (Easterbrook, Singer, Storey, & Damian, 2008). Thus, the following hypotheses were defined, followed by their respective null

hypotheses, those that researchers aim to refute.

**H1**. *It is possible to model the SM performance based on the interactions of users on the official profiles of candidates and find correlations between the SM and the electoral performances of candidates.*

**H1'**. *It is not possible to model the SM performance based on the interactions of users on the official profiles of candidates and find correlations between the SM and the electoral performances of candidates.*

**H2'**. *There are patterns among different countries between the SM performance of candidates and electoral results.*

**H2'**. *It is not possible to find patterns among different countries between the SM performance of candidates and electoral results.*

The methodology for rejecting $H_1$' consists of three steps:

A. To reuse Brito's concept of SM performance (Brito, Meira, & Adeodato, 2021) by defining and detailing the specific metrics related to performance;
B. To collect the performance metrics on SM platforms and the final vote share of candidates from studied elections;
C. To analyze the data by running Pearson correlation tests for correlations between the SM performance metrics and the electoral performance.

The methodology for rejecting $H_2$' consists of an exploratory analysis on the results of $H_1$', in order to verify whether similar correlations are presented among different countries.

For the experiments, we chose countries throughout Latin America because they present a similar context: they are situated in the same region, they have similar historical origins, languages, characteristics, and electoral procedures, such as the direct vote, the two-round system, many candidates running in the first round, and there is unequal exposure on TV and in the traditional media. Thus, geographical, political, or other unintentional bias was reduced. Hence, presidential elections held in Colombia, Mexico, and Brazil in 2018 and Argentina in 2019 were chosen. These are the four most populous countries in Latin America, home to 70% of the population (United Nations, 2019), and were responsible for 81% of the GDP in 2019 (The World Bank, 2022), which increases the validity of our experiments. We only considered the first round of elections because the election context is different in the second round in which there are only two candidates, who have the same exposure on TV and in the traditional media.

The most used newsfeed-based SM media platform in these countries are Youtube, Facebook, Instagram, and Twitter (Kemp, 2020a, 2020b, 2020c, 2020d). Youtube data were not considered in this study because we identified that at the time of elections, many candidates did not have an official profile on this platform, or its use was irregular.

### 2.1. Measuring SM performance

The measurement of SM performance applied in this study uses the approach presented by Brito et al. (Brito, Meira, & Adeodato, 2021), inspired by Zajonc's exposure theory (Zajonc, 1968), in conjunction with an understanding of the functioning of the algorithms in social media platforms.

In 1968 (Zajonc, 1968) and beyond (Murphy & Zajonc, 1993; Zajonc, 1980, 2001), Zajonc's studies hypothesized that "mere repeated exposure of the individual to a stimulus object enhances his attitude towards it." This effect, also termed the familiarity principle, has been demonstrated in many different contexts and also applies to affective reactions (Swap, 1977), when it may be considered that people tend to have better attitudes towards others whom they are used to seeing. Applying these theories in the electoral context, in 1986, Oppenheimer (Oppenheimer, Stimson, & Waterman, 1986) reported a correlation between the exposure of politicians and their electoral performance, and Mondak (Mondak, 1995) observed that "media exposure fuels political discussion."

Considering this effect, the performance measurement focuses on analyzing the number of people who pay attention to the candidates by interacting with their content within the context of SM. For this, we considered the interactions on posts of the official profiles of candidates on newsfeed platforms, such as Facebook, Twitter, and Instagram.

Interactions on the candidates' posts consist of the likes, comments, and shares on each post. These actions indicate that the user has seen and paid attention to the content and actively acted. One like may be considered a quick, easy endorsement of the content; a comment demands more cognitive effort and may be positive or negative; and a share replicates the content to the user's own network, thereby actively helping to propagate it. In the case of Facebook, a like has subtypes, such as "Like", "Love", "Haha", "Wow", "Sad" and "Angry". However, there are no distinctions among these interactions in practice, which may be considered just one. This is because even negative reactions, such as "Sad" and "Angry", are usually negative regarding the content of the post, such as the reporting of a sad situation, and not a disagreement with whoever posted it.

Indeed, all these actions, not only by sharing, help to propagate a candidate's presence online. As SM algorithms prioritize showing the content of users with more engagement (Lars Backstrom & The Facebook, 2013; Youtube Inc., 2017), this creates a snowball effect. As more people interact with a post, so it is shown to more people, leading to more people interacting with it. The end result of the exposure theory is that more engagement and more exposure may be correlated with a better attitude towards a candidate and more votes.

As a consequence, regardless of the social network under analysis, we have considered the metrics related to the number of likes, comments, and shares (or similar terms, such as retweets on Twitter as a synonym for shares) received by the official posts of candidates. We have considered the absolute numbers in a period and the relative numbers per post.

Let $p_c(d)$ denote the number of posts made by a candidate $c$ on an arbitrary SM platform (such as Facebook) on day $d$, and $l(p_c(d))$ denote the total number of likes received by the posts made by the candidate $c$ on day $d$. Similarly, let $s(p_c(d))$ denote the total number of shares received by the posts of the candidate, and $cm(pc(d))$ the total number of comments received by the posts of the candidate $c$ on day $d$. Thus, given a time window from $d = i$ to $d = f$, we consider the following metrics:

The number of posts $P$ made by the candidate $c$ on the SM platform within the time window:

$$P = \sum_{d=i}^{d=f} p_c(d)$$

The number of likes $L$ received by the posts of the candidate $c$ on the SM platform within the time window:

$$L = \sum_{d=i}^{d=f} l(p_c(d))$$

The number of shares $S$ received by the posts of the candidate $c$ on the SM platform within the time window:

$$S = \sum_{d=i}^{d=f} s(p_c(d))$$

The number of comments $CM$ received by the posts of the candidate $c$ on the SM platform within the time window:

$$CM = \sum_{d=i}^{d=f} cm(p_c(d))$$

The average number of likes per post ($LP$), shares per post ($SP$) and comments per post ($CMP$) received by the posts of the candidate $c$ on the SM platform within the time window:

$$LP = L/P$$

$$SP = S/P$$

$$CMP = CM/_P$$

These metrics are generic and well suited for most newsfeed-based SM platforms. In the specific case of this study, we consider Facebook, Twitter, and Instagram as SM platforms, as justified at the beginning of Section 2. However, if other relevant platforms are created or identified as being relevant in other election scenarios, their metrics may also be added by following the same rationale of interactions, even if they present slight differences. For example, in the case of YouTube, the number of visualizations, likes, dislikes and comments on the videos posted by the candidates should be considered. Table 1 summarizes the metrics used.

### 2.2. Data collection

An information system was developed and run from 2016, entirely focused on collecting and analyzing daily SM data. It has passed the verification process for accessing to Facebook/Instagram and Twitter APIs, according to the official guidelines of each platform (Facebook Inc., 2020; Twitter Inc., 2020).

To avoid bias generated by candidates with very small vote intentions, a concern pointed out since the beginning of this research area (Jungherr et al., 2012), the study only considered candidates with more than 1% of vote share. The list of candidates is presented in Table 2. Also, to avoid bias regarding the arbitrary selection of a unique time frame, another frequent concern in this subject (Brito, Silva Filho, & Adeodato, 2021; Jungherr et al., 2012), data collection began 10 months (300 days) before the elections, when the candidatures start to take form. The election timelines, including the election date, the campaign launch, and the start of data collection, are presented in Table 3.

The candidate profiles on these three networks are presented in Table 4. Those marked with a star represent unverified profiles at the time of collection. At that time, Instagram profiles of the Mexican candidates Obrador and Cortés were personal profiles rather than commercial profiles. This setup does not allow for data to be collected through the Instagram API. Thus, Instagram data on these candidates were not used in this study.

Lastly, the final vote share of candidates was gathered from the official electoral courts.

### 2.3. Data analysis

This study was performed after the elections. Therefore, to avoid bias related to the arbitrary selection of the data collection period, the correlation was tested with data regarding many different periods before election day. Thus, 20 different time windows were considered: from 1 to 7 days, from 1 to 4 weeks, and from 1 to 10 months ($w = 1, 2, 3, 4, 5, 6, 7, 14, 21, 28, 30, 60, 90, 120, 150, 180, 210, 240, 270, 300$ days). We subsequently analyzed the SM performance of candidates during $w$ days leading up to election day.

We performed the analysis by calculating the Pearson correlation coefficient (r), a well-known metric for covariance measure, to calculate the correlations between all the defined SM performance metrics and the number of votes received. The coefficient ($r$) ranges from $-1$ to 1. A value of 1 implies that a linear equation perfectly describes the relationship between the two variables, with all data points lying on a line for which one variable increases as the other also increases. A value of $-1$ implies a perfect negative correlation. A value of 0 implies that there is no linear correlation between the variables. Details of calculations can be found in Chapter 7 of (Boslaugh, 2012). There are many guidelines for the interpretation of $r$, such as (Goodwin & Leech, 2006), although it is dependent on context: a correlation of 0.8 is usually considered a high correlation, especially in the social sciences. However, the same correlation may be low if a physical law is verified using high-quality instruments in controlled experiments. Also, the small number of samples, as in the case of this study, may interfere with the interpretation. Thus, even though it is commonly accepted that an $r \geq 0.7$ indicates a high correlation, in this study we use the correlation coefficient rule of thumb, statistically justified in (Krehbiel, 2004). Considering $r$ as the Pearson correlation coefficient and $n$ the number of samples, three rules are presented:

- Rule of Thumb No. 1: If $|r_{xy}| \geq 2/\sqrt{n}$, then a linear relationship exists.
- Rule of Thumb No. 2: If $|r_{xy}| \geq 2/\sqrt{n+1}$, then a linear relationship exists.
- Rule of Thumb No. 3: If $|r_{xy}| \geq 2/\sqrt{n+2}$, then a linear relationship exists.

As it is recognized by (Krehbiel, 2004) that rule No. 1 is slightly conservative, to reject $H_1$' we will consider the threshold of rule No. 3 as the threshold indicating a correlation, the threshold of rule No. 2 as indicating a high correlation, and the threshold of rule No. 1 as indicating a very high correlation. Thus, to reject $H_1$', at least one defined metric must present a correlation with the electoral results.

In addition, to reject $H_2$', we will verify if similar correlations occur

**Table 1**
Metrics for performance measurement (Brito, Meira, & Adeodato, 2021).

| Social network | Metric | Description |
|---|---|---|
| Facebook | FBPosts | Sum of posts in the period |
| | FBLikes | Sum of likes in the period |
| | FBShares | Sum of shares in the period |
| | FBComments | Sum of comments in the period |
| | FBLikesPPost | Average of likes per post in the period |
| | FBSharesPPost | Average of shares per post in the period |
| | FBCommentsPPost | Average of comments per post in the period |
| Twitter | TTPosts | Sum of posts in the period |
| | TTLikes | Sum of likes in the period |
| | TTRetweets | Sum of retweets in the period |
| | TTLikesPPost | Average of likes per post in the period |
| | TTRetweetsPPost | Average of retweets per post in the period |
| Instagram | IGPosts | Sum of posts in the period |
| | IGLikes | Sum of likes in the period |
| | IGComments | Sum of comments in the period |
| | IGLikesPPost | Average of likes per post in the period |
| | IGCommentsPPost | Average of comments per post in the period |

**Table 2**
List of candidates.

| Argentina | Brazil | Colombia | Mexico |
|---|---|---|---|
| Alberto Fernandez | Jair Bolsonaro | Iván Duque Márquez | Andrés Manuel López Obrador |
| Mauricio Macri | Fernando Haddad | Gustavo Petro | Ricardo Anaya Cortés |
| Roberto Lavagna | Ciro Gomes | Sergio Fajardo | José Antonio Meade Kuribreña |
| Nicolas del Caño | Geraldo Alckmin | Germán Vargas Lleras | Jaime Rodriguez Calderon |
| Juan José Gómez Centurión | João Amoêdo | Humberto De la Calle | |
| José Luis Espert | | | |
| *6 Candidates, all considered* | *13 candidates, 5 considered* | *8 Candidates, 5 considered* | *4 candidates, all considered* |

**Table 3**
Election timelines.

| Country | Election date | Campaign launch | Start of data collection |
|---|---|---|---|
| Argentina | 27/10/2019 | 07/07/2019 | 31/12/2018 |
| Brazil | 07/10/2018 | 16/08/2018 | 11/12/2017 |
| Colombia | 27/05/2018 | 27/01/2018 | 31/07/2017 |
| Mexico | 01/07/2018 | 30/03/2018 | 04/09/2017 |

*Date format: dd/mm/yyyy.

**Table 4**
Candidate profiles on facebook, twitter, and instagram.

| Candidate - Argentina | Facebook | Twitter | Instagram |
|---|---|---|---|
| Alberto Fernandez | alferdezok | alferdez | alferdezok |
| Mauricio Macri | mauriciomacri | mauriciomacri | mauriciomacri |
| Roberto Lavagna | LavagnaARG | Rlavagna | rlavagna |
| Nicolas del Caño | NicolasDelCano.PTS | NicolasdelCano | nico_del_cano |
| Juan José Gómez Centurión | juanjomalvinas | juanjomalvinas[a] | – |
| José Luis Espert | JLEspert[a] | jlespert | joseluisespert |
| | | | |
| Candidate - Brazil | Facebook | Twitter | Instagram |
| Jair Bolsonaro | jairmessias.bolsonaro | jairbolsonaro | jairmessiasbolsonaro |
| Fernando Haddad | fernandohaddad | Haddad_Fernando | fernandohaddadoficial |
| Ciro Gomes | cirogomesoficial | cirogomes | cirogomes |
| Geraldo Alckmin | geraldoalckmin | geraldoalckmin | geraldoalckmin_ |
| João Amoêdo | JoaoAmoedoNOVO | joaoamoedonovo | joaoamoedonovo |
| | | | |
| Candidate - Colombia | FB | TT | IG |
| Iván Duque Márquez | ivanduquemarquez | IvanDuque | ivanduquemarquez |
| Gustavo Petro | gustavopetrourrego | petrogustavo | gustavopetrourrego |
| Sergio Fajardo | SergioFajardoV | sergio_fajardo | sergiofajardovalderrama |
| Germán Vargas Lleras | GermanVargasLleras | German_Vargas | germanvargaslleras[a] |
| Humberto De la Calle | DeLaCalleHum | DeLaCalleHum | delacallehum[a] |
| | | | |
| Candidate - Mexico | Facebook | Twitter | Instagram |
| Andrés Manuel López Obrador | lopezobrador.org.mx | lopezobrador_ | [b] |
| Ricardo Anaya Cortés | RicardoAnayaC | RicardoAnayaC | [b] |
| José Antonio Meade Kuribreña | JoseAMeadeK | JoseAMeadeK | joseameadek |
| Jaime Rodriguez Calderon | JaimeRodriguezElBronco | JaimeRdzNL | jaimerodriguezcalderon |

[a] Unverified profiles.
[b] Not commercial profiles at the period of data collection. Thus, data collection on these profiles was not possible.

among different countries and if the same metrics present the highest correlations in many countries.

## 3. Results and discussion

### 3.1. Overview of collected data

A total of 67,403 posts and 449,201,837 interactions were collected from the candidate profiles in all four countries, which makes an average of 6664 interactions per post. Tables 5–8 presents an overview of the collected data from the posts of candidates in each country.

A total of 15,432 posts was collected from the Argentinian candidates: 3224 from Facebook, 10,444 from Twitter, and 1764 from Instagram. Although the majority were posted on Twitter, this network received the lowest total number of interactions, both raw interactions and the number of interactions per post. On the other hand, the lowest number of posts were on Instagram, although it presented the highest mean and median interactions per post.

**Table 5**
Overview of the collected data from the posts of Argentinian candidates.

| Posts - Argentina | Total | Min | Max | Mean | Median | Std. dev. |
|---|---|---|---|---|---|---|
| Facebook Posts | 3224 | – | – | – | – | – |
| Facebook Likes | 23,550,969 | 0 | 361,940 | 7305 | 746 | 17,270 |
| Facebook Comments | 4,609,286 | 0 | 97,234 | 1430 | 99 | 4148 |
| Facebook Shares | 5,155,555 | 0 | 54,282 | 1599 | 220 | 3835 |
| Twitter Posts | 10,444 | – | – | – | – | – |
| Twitter Likes | 15,427,184 | 0 | 123,390 | 1477 | 46 | 4722 |
| Twitter Shares | 4,454,052 | 0 | 43,386 | 426 | 18 | 1346 |
| Instagram Posts | 1764 | – | – | – | – | – |
| Instagram Likes | 28,249,171 | 13 | 267,786 | 16,014 | 7515 | 23,513 |
| Instagram Comments | 1,353,965 | 0 | 32,296 | 768 | 276 | 1764 |

A total of 19,586 posts was collected from the Brazilian candidates: 6101 from Facebook, 10,181 from Twitter, and 3304 from Instagram. As in Argentina, the majority were posted on Twitter, although this network received the lowest number of interactions. Instagram also received the lowest number of posts but the highest mean and median interactions.

A total of 22,542 posts was collected from the Colombian candidates: 3746 from Facebook, 15,996 from Twitter and 2800 from Instagram. As in Argentina and Brazil, the majority of posts were made on Twitter, but this network received the lowest number of interactions. However, a different result was obtained regarding the mean and median of interactions per post, as it was obtained by posts on Facebook.

Lastly, a total of 9843 posts was collected from the Mexican candidates: 2308 from Facebook, 7146 from Twitter, and only 389 from Instagram. As previously mentioned, we did not collect data from Instagram accounts of the two most voted candidates because these were personal profiles rather than commercial, and it impacted the analysis. As in other countries, the majority were posted on Twitter, but the highest interactions were on Facebook. Even only collecting data for the 3rd and 4th most voted candidates, their interactions per post on Instagram were higher than the interaction per post of all the candidates on Twitter.

To summarize, candidates in all countries mostly posted on Twitter, but the public response on this platform is lower than the response on Facebook and Instagram. In Argentina and Brazil, most interactions were on Instagram, while in Colombia, most interactions were on Facebook. Lastly, there was a large variation in the interactions on posts since they are able to receive between zero and one million interactions, and there is a large standard deviation in all metrics.

The total of 67,403 collected posts from the candidate profiles in all four countries makes an average of 16,851 per country. This is the smallest amount of collected data if compared to the usual approaches: (Brito, Silva Filho, & Adeodato, 2021) identified an average of 12.9

**Table 6**
Overview of the collected data from the posts of Brazilian candidates.

| Posts - Brazil | Total | Min | Max | Mean | Median | Std. dev. |
|---|---|---|---|---|---|---|
| Facebook Posts | 6101 | – | – | – | – | – |
| Facebook Likes | 81,741,405 | 1 | 726,899 | 13,398 | 4084 | 29,881 |
| Facebook Comments | 12,602,311 | 0 | 798,554 | 2066 | 417 | 17,403 |
| Facebook Shares | 27,192,425 | 0 | 359,513 | 4457 | 1047 | 14,324 |
| Twitter Posts | 10,181 | – | – | – | – | – |
| Twitter Likes | 21,330,941 | 0 | 107,368 | 2095 | 358 | 6125 |
| Twitter Shares | 5,493,573 | 0 | 62,105 | 540 | 110 | 1560 |
| Instagram Posts | 3304 | – | – | – | – | – |
| Instagram Likes | 101,964,341 | 309 | 1,190,258 | 30,861 | 6531 | 84,142 |
| Instagram Comments | 3,444,310 | 0 | 91,469 | 1042 | 189 | 4032 |

**Table 7**
Overview of the collected data from the posts of Colombian candidates.

| Posts - Colombia | Total | Min | Max | Mean | Median | Std. dev. |
|---|---|---|---|---|---|---|
| Facebook Posts | 3746 | – | – | – | – | – |
| Facebook Likes | 14,548,883 | 5 | 117,504 | 3884 | 945 | 8566 |
| Facebook Comments | 3,488,095 | 0 | 59,747 | 931 | 115 | 2953 |
| Facebook Shares | 5,485,706 | 0 | 171,825 | 1464 | 206 | 5135 |
| Twitter Posts | 15,996 | – | – | – | – | – |
| Twitter Likes | 10,865,730 | 0 | 35,230 | 679 | 200 | 1467 |
| Twitter Shares | 4,635,743 | 0 | 12,540 | 290 | 103 | 567 |
| Instagram Posts | 2800 | – | – | – | – | – |
| Instagram Likes | 5,052,473 | 28 | 32,534 | 1804 | 712 | 2868 |
| Instagram Comments | 134,758 | 0 | 1694 | 48 | 20 | 96 |

**Table 8**
Overview of the collected data from the posts of Mexican candidates.

| Posts - Mexico | Total | Min | Max | Mean | Median | Std. dev. |
|---|---|---|---|---|---|---|
| Facebook Posts | 2308 | – | – | – | – | – |
| Facebook Likes | 38,046,763 | 15 | 357,567 | 16,485 | 5605 | 30,696 |
| Facebook Comments | 7,366,475 | 1 | 132,499 | 3192 | 945 | 7821 |
| Facebook Shares | 10,440,845 | 0 | 165,399 | 4524 | 1469 | 10,652 |
| Twitter Posts | 7146 | – | – | – | – | – |
| Twitter Likes | 8,081,106 | 0 | 35,744 | 1131 | 174 | 2731 |
| Twitter Shares | 3,585,676 | 0 | 15,191 | 502 | 71 | 1100 |
| Instagram Posts | 389 | – | – | – | – | – |
| Instagram Likes | 870,331 | 127 | 16,207 | 2237 | 1293 | 2417 |
| Instagram Comments | 29,765 | 1 | 719 | 77 | 39 | 101 |

million and a median of 250 thousand datapoints (usually tweets) in current studies for a single election. If we consider the median, we have collected 17,509 posts per country, which is only 7.0% of the data collected by other studies. Moreover, all the required data was collected over a long period (10 months), instead of just a sample of tweets during arbitrary periods, as obtained in the most commonly used approach. Despite the much smaller amount of collected data, this paper's data is better spread along time and the performance metrics used are more meaningful, as will be confirmed in the next sections.

In order to understand voter engagement on SM candidate profiles in the studied countries, we compared the interaction numbers with the total population in each country in 2019 (United Nations, 2019). To avoid bias in the analysis, because of the lack of data on Instagram for two profiles of Mexican candidates, we created two summaries: one containing all the collected data, and another containing data only from Facebook and Twitter. Results are presented in Table 9 and demonstrate that the Argentinians were more engaged with the profiles of the candidates and the citizens from Mexico presented the lowest level of engagement, regardless of including Instagram data.

### 3.2. Correlations results

As defined in Section 2.3, we calculated the Pearson correlation using three different thresholds based on the number of samples (Krehbiel, 2004).

For Argentina, six candidates were considered. Thus, the adopted thresholds were $r \geq 0.71$ for showing a correlation, $r \geq 0.76$ for a high correlation, and $r \geq 0.82$ for a very high correlation. Table 10 presents the results. From Tables 10–13, cells in dark gray highlight the values above the highest threshold, and light gray highlights the values higher than the lowest threshold. The last column presents a simple average of the results from all windows, which is the sorting criterion of the tables.

The data reveals that many of the defined variables presented a very high correlation with the results of the Argentinian elections, with emphasis on the variables related to the number of interactions per post on all three networks. Higher correlations were concentrated in the windows between 21 and 30 days, and changes smoothly between nearby windows, as expected. In opposition to this, the absolute number of posts on all three networks presented the lowest correlations, and the number of tweets presented a negative correlation. Moreover, the correlations considering the window of 1 day were notably lower than the other windows.

For Brazil and Colombia, five candidates were considered. Thus, the adopted thresholds were $r \geq 0.76$ for showing a correlation, $r \geq 0.82$ for a high correlation, and $r \geq 0.89$ for a very high correlation. Tables 11 and 12 present the results for Brazil and Colombia, respectively.

In the Brazilian scenario, although there was no correlation between

**Table 9**
Comparison of engagement on SM and the total population.

| All platforms | | | | |
|---|---|---|---|---|
| Country | Argentina | Brazil | Colombia | Mexico |
| Population | 44,781,000 | 211,050,000 | 50,339,000 | 127,576,000 |
| Total Posts | 15,432 | 19,586 | 22,542 | 9843 |
| Interactions | 82,800,182 | 253,769,306 | 44,211,388 | 68,420,961 |
| Interactions/Post | 5365 | 12,957 | 1961 | 6951 |
| *Interactions/ Habitant* | *1.85* | *1.20* | *0.88* | *0.54* |
| **Only Facebook and Twitter** | | | | |
| Country | Argentina | Brazil | Colombia | Mexico |
| Population | 44,781,000 | 211,050,000 | 50,339,000 | 127,576,000 |
| Total Posts | 13,668 | 16,282 | 19,742 | 9454 |
| Interactions | 53,197,046 | 148,360,655 | 39,024,157 | 67,520,865 |
| Interactions/Post | 3892 | 9112 | 1977 | 7142 |
| *Interactions/ Habitant* | *1.19* | *0.70* | *0.78* | *0.53* |

**Table 10**
Pearson correlation results for the SM performance and the Argentinian election results.

| Metric - Argentina | Number of Days | | | | | | | | | | | | | | | | | | | | Avg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 14 | 21 | 28 | 30 | 60 | 90 | 120 | 150 | 180 | 210 | 240 | 270 | 300 | |
| IGLikesPPost | 0.62 | 0.87 | 0.94 | 0.91 | 0.90 | 0.94 | 0.94 | 0.95 | 0.95 | 0.94 | 0.94 | 0.92 | 0.91 | 0.90 | 0.91 | 0.92 | 0.93 | 0.93 | 0.93 | 0.93 | 0.91 |
| TTSharesPPost | 0.71 | 0.80 | 0.94 | 0.93 | 0.92 | 0.83 | 0.82 | 0.92 | 0.93 | 0.95 | 0.93 | 0.93 | 0.90 | 0.91 | 0.92 | 0.86 | 0.81 | 0.76 | 0.73 | 0.69 | 0.86 |
| FBSharesPPost | 0.66 | 0.68 | 0.73 | 0.78 | 0.81 | 0.87 | 0.87 | 0.91 | 0.95 | 0.95 | 0.95 | 0.95 | 0.92 | 0.93 | 0.93 | 0.85 | 0.85 | 0.84 | 0.84 | 0.82 | 0.85 |
| TTLikesPPost | 0.70 | 0.82 | 0.97 | 0.95 | 0.94 | 0.81 | 0.81 | 0.91 | 0.92 | 0.94 | 0.92 | 0.92 | 0.88 | 0.88 | 0.90 | 0.84 | 0.79 | 0.74 | 0.71 | 0.67 | 0.85 |
| FBLikesPPost | 0.65 | 0.64 | 0.69 | 0.70 | 0.72 | 0.80 | 0.80 | 0.89 | 0.93 | 0.91 | 0.90 | 0.90 | 0.86 | 0.88 | 0.88 | 0.81 | 0.81 | 0.81 | 0.80 | 0.79 | 0.81 |
| IGCommentsPPost | 0.36 | 0.78 | 0.85 | 0.82 | 0.83 | 0.91 | 0.92 | 0.90 | 0.91 | 0.88 | 0.87 | 0.81 | 0.77 | 0.75 | 0.76 | 0.78 | 0.78 | 0.78 | 0.78 | 0.77 | 0.80 |
| TTShares | 0.68 | 0.85 | 0.71 | 0.76 | 0.76 | 0.72 | 0.72 | 0.75 | 0.76 | 0.76 | 0.76 | 0.79 | 0.86 | 0.90 | 0.90 | 0.88 | 0.87 | 0.86 | 0.85 | 0.83 | 0.80 |
| TTLikes | 0.65 | 0.85 | 0.75 | 0.78 | 0.77 | 0.69 | 0.70 | 0.74 | 0.75 | 0.75 | 0.74 | 0.78 | 0.84 | 0.87 | 0.87 | 0.85 | 0.84 | 0.83 | 0.82 | 0.80 | 0.78 |
| FBCommentsPPost | 0.11 | 0.61 | 0.72 | 0.70 | 0.74 | 0.86 | 0.87 | 0.90 | 0.91 | 0.89 | 0.88 | 0.86 | 0.79 | 0.82 | 0.82 | 0.75 | 0.75 | 0.74 | 0.73 | 0.72 | 0.76 |
| FBShares | 0.66 | 0.65 | 0.62 | 0.67 | 0.70 | 0.75 | 0.74 | 0.75 | 0.75 | 0.76 | 0.75 | 0.77 | 0.77 | 0.78 | 0.77 | 0.74 | 0.74 | 0.73 | 0.72 | 0.72 | 0.73 |
| IGLikes | 0.62 | 0.87 | 0.74 | 0.74 | 0.72 | 0.70 | 0.68 | 0.69 | 0.68 | 0.69 | 0.68 | 0.70 | 0.74 | 0.75 | 0.75 | 0.73 | 0.72 | 0.71 | 0.70 | 0.70 | 0.72 |
| FBLikes | 0.65 | 0.63 | 0.60 | 0.62 | 0.64 | 0.70 | 0.70 | 0.73 | 0.73 | 0.73 | 0.71 | 0.73 | 0.72 | 0.74 | 0.73 | 0.71 | 0.71 | 0.70 | 0.70 | 0.70 | 0.69 |
| FBComments | 0.11 | 0.61 | 0.62 | 0.63 | 0.65 | 0.75 | 0.74 | 0.74 | 0.72 | 0.71 | 0.70 | 0.70 | 0.68 | 0.70 | 0.69 | 0.67 | 0.67 | 0.66 | 0.65 | 0.65 | 0.65 |
| IGComments | 0.36 | 0.78 | 0.67 | 0.67 | 0.67 | 0.68 | 0.67 | 0.66 | 0.65 | 0.65 | 0.64 | 0.64 | 0.64 | 0.65 | 0.65 | 0.64 | 0.63 | 0.62 | 0.62 | 0.61 | 0.64 |
| IGPosts | 0.03 | 0.55 | 0.41 | 0.44 | 0.45 | 0.45 | 0.43 | 0.51 | 0.51 | 0.57 | 0.57 | 0.59 | 0.60 | 0.63 | 0.58 | 0.50 | 0.46 | 0.45 | 0.43 | 0.42 | 0.48 |
| FBPosts | 0.29 | 0.80 | 0.71 | 0.72 | 0.71 | 0.61 | 0.58 | 0.27 | 0.29 | 0.37 | 0.40 | 0.26 | 0.14 | 0.13 | 0.13 | 0.27 | 0.24 | 0.30 | 0.32 | 0.37 | 0.40 |
| TTPosts | -0.36 | -0.41 | -0.04 | -0.18 | -0.37 | -0.35 | -0.36 | -0.24 | -0.13 | -0.06 | -0.06 | -0.32 | -0.43 | -0.47 | -0.53 | -0.51 | -0.47 | -0.38 | -0.33 | -0.23 | -0.31 |

**Table 11**
Pearson correlation results for the SM performance and the Brazilian election results.

| Metric - Brazil | Number of Days | | | | | | | | | | | | | | | | | | | | Avg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 14 | 21 | 28 | 30 | 60 | 90 | 120 | 150 | 180 | 210 | 240 | 270 | 300 | |
| FBComments | 0.92 | 0.94 | 0.89 | 0.89 | 0.89 | 0.88 | 0.88 | 0.90 | 0.91 | 0.92 | 0.92 | 0.91 | 0.90 | 0.90 | 0.89 | 0.89 | 0.88 | 0.88 | 0.88 | 0.87 | 0.90 |
| IGLikes | 0.87 | 0.85 | 0.86 | 0.87 | 0.87 | 0.87 | 0.87 | 0.87 | 0.88 | 0.89 | 0.89 | 0.88 | 0.88 | 0.87 | 0.87 | 0.87 | 0.87 | 0.87 | 0.87 | 0.87 | 0.87 |
| IGComments | 0.83 | 0.87 | 0.87 | 0.88 | 0.88 | 0.88 | 0.88 | 0.87 | 0.88 | 0.88 | 0.88 | 0.87 | 0.87 | 0.87 | 0.86 | 0.86 | 0.86 | 0.86 | 0.86 | 0.86 | 0.87 |
| IGLikesPPost | 0.86 | 0.85 | 0.85 | 0.85 | 0.84 | 0.84 | 0.84 | 0.84 | 0.85 | 0.85 | 0.85 | 0.85 | 0.85 | 0.86 | 0.87 | 0.87 | 0.88 | 0.88 | 0.88 | 0.88 | 0.86 |
| TTShares | 0.86 | 0.77 | 0.80 | 0.87 | 0.86 | 0.87 | 0.88 | 0.87 | 0.88 | 0.88 | 0.88 | 0.87 | 0.87 | 0.86 | 0.86 | 0.85 | 0.85 | 0.85 | 0.84 | 0.84 | 0.86 |
| FBCommentsPPost | 0.84 | 0.85 | 0.84 | 0.84 | 0.84 | 0.84 | 0.84 | 0.84 | 0.84 | 0.85 | 0.85 | 0.85 | 0.85 | 0.86 | 0.84 | 0.84 | 0.85 | 0.85 | 0.85 | 0.85 | 0.85 |
| FBSharesPPost | 0.87 | 0.85 | 0.85 | 0.86 | 0.86 | 0.86 | 0.86 | 0.85 | 0.85 | 0.85 | 0.85 | 0.83 | 0.83 | 0.84 | 0.81 | 0.81 | 0.81 | 0.82 | 0.81 | 0.82 | 0.84 |
| FBLikesPPost | 0.85 | 0.85 | 0.85 | 0.85 | 0.85 | 0.85 | 0.85 | 0.84 | 0.83 | 0.84 | 0.84 | 0.82 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.84 | 0.84 | 0.84 |
| IGCommentsPPost | 0.87 | 0.85 | 0.84 | 0.84 | 0.83 | 0.82 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 |
| TTLikes | 0.80 | 0.70 | 0.74 | 0.82 | 0.83 | 0.83 | 0.84 | 0.85 | 0.86 | 0.87 | 0.87 | 0.86 | 0.86 | 0.86 | 0.85 | 0.85 | 0.84 | 0.84 | 0.84 | 0.84 | 0.83 |
| FBShares | 0.97 | 0.87 | 0.91 | 0.92 | 0.93 | 0.92 | 0.92 | 0.88 | 0.88 | 0.87 | 0.86 | 0.74 | 0.74 | 0.73 | 0.73 | 0.74 | 0.74 | 0.74 | 0.75 | 0.75 | 0.83 |
| FBLikes | 0.93 | 0.90 | 0.91 | 0.90 | 0.90 | 0.88 | 0.88 | 0.84 | 0.84 | 0.84 | 0.83 | 0.74 | 0.74 | 0.74 | 0.74 | 0.75 | 0.75 | 0.75 | 0.76 | 0.76 | 0.82 |
| TTLikesPPost | 0.77 | 0.83 | 0.83 | 0.84 | 0.84 | 0.79 | 0.79 | 0.80 | 0.81 | 0.82 | 0.82 | 0.82 | 0.81 | 0.80 | 0.80 | 0.80 | 0.79 | 0.79 | 0.78 | 0.77 | 0.81 |
| TTSharesPPost | 0.76 | 0.81 | 0.82 | 0.83 | 0.83 | 0.78 | 0.79 | 0.79 | 0.80 | 0.81 | 0.81 | 0.81 | 0.80 | 0.79 | 0.78 | 0.78 | 0.78 | 0.77 | 0.76 | 0.76 | 0.79 |
| FBPosts | 0.34 | 0.21 | 0.19 | 0.22 | 0.23 | 0.23 | 0.24 | 0.21 | 0.16 | 0.11 | 0.10 | 0.08 | 0.04 | 0.00 | 0.00 | 0.00 | -0.02 | -0.02 | 0.00 | 0.00 | 0.12 |
| TTPosts | 0.41 | 0.17 | 0.14 | 0.18 | 0.15 | 0.16 | 0.18 | 0.03 | -0.01 | -0.07 | -0.07 | -0.09 | -0.08 | -0.08 | -0.08 | -0.10 | -0.11 | -0.11 | -0.11 | -0.10 | 0.02 |
| IGPosts | 0.12 | 0.00 | 0.00 | 0.06 | 0.07 | 0.08 | 0.10 | 0.08 | 0.06 | 0.03 | 0.03 | 0.00 | -0.03 | -0.05 | -0.06 | -0.06 | -0.07 | -0.07 | -0.06 | -0.05 | 0.01 |

the number of posts and votes, all the other variables presented some correlation on most windows, and most variables presented at least a high correlation. However, a clear pattern could not be observed.

The performance of the SM variables regarding the Colombian elections was lower than the results for Argentina and Brazil. The highest correlations were related to comments, both the absolute number as well as the number averaged per post, on Facebook and Instagram. The lowest correlations were related to Twitter, and a high negative correlation was related to the absolute number of posts on Facebook.

For Mexico, only four candidates were considered. Thus, the adopted thresholds were $r \geq 0.82$ for showing a correlation, and $r \geq 0.89$ for a high correlation. The third threshold may only be achievable if $r = 1.0$. Table 13 presents the results.

As in the case of Argentina, the number of interactions per post presented the highest correlations in all windows, close to one, especially when considering the Twitter platform. In agreement with results in other countries, the total number of posts presented a negative correlation. As the Instagram data for Mexican candidates is incomplete, it was not considered and has been struck through in Table 13.

**Table 12**
Pearson correlation results for the SM performance and the Colombian election results.

| Metric - Colombia | Number of Days | | | | | | | | | | | | | | | | | | | | Avg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 14 | 21 | 28 | 30 | 60 | 90 | 120 | 150 | 180 | 210 | 240 | 270 | 300 | |
| FBComments | 0.72 | 0.84 | 0.91 | 0.82 | 0.83 | 0.88 | 0.96 | 0.83 | 0.86 | 0.88 | 0.87 | 0.87 | 0.86 | 0.80 | 0.80 | 0.80 | 0.79 | 0.79 | 0.79 | 0.77 | 0.83 |
| FBCommentsPPost | 0.76 | 0.79 | 0.81 | 0.83 | 0.83 | 0.86 | 0.90 | 0.92 | 0.92 | 0.91 | 0.91 | 0.90 | 0.88 | 0.84 | 0.83 | 0.80 | 0.76 | 0.72 | 0.69 | 0.65 | 0.83 |
| IGComments | 0.35 | 0.35 | 0.47 | 0.56 | 0.61 | 0.73 | 0.79 | 0.88 | 0.94 | 0.94 | 0.94 | 0.94 | 0.94 | 0.93 | 0.93 | 0.93 | 0.92 | 0.92 | 0.92 | 0.92 | 0.80 |
| IGCommentsPPost | 0.72 | 0.79 | 0.89 | 0.92 | 0.94 | 0.95 | 0.92 | 0.96 | 0.90 | 0.84 | 0.85 | 0.82 | 0.78 | 0.72 | 0.66 | 0.62 | 0.58 | 0.55 | 0.51 | 0.47 | 0.77 |
| IGLikesPPost | 0.52 | 0.58 | 0.66 | 0.69 | 0.72 | 0.76 | 0.78 | 0.81 | 0.81 | 0.79 | 0.82 | 0.90 | 0.90 | 0.88 | 0.83 | 0.79 | 0.77 | 0.73 | 0.69 | 0.64 | 0.75 |
| IGLikes | 0.29 | 0.28 | 0.35 | 0.38 | 0.40 | 0.48 | 0.52 | 0.62 | 0.77 | 0.82 | 0.84 | 0.88 | 0.92 | 0.93 | 0.92 | 0.92 | 0.91 | 0.91 | 0.91 | 0.91 | 0.70 |
| FBLikesPPost | 0.99 | 0.91 | 0.77 | 0.71 | 0.71 | 0.63 | 0.61 | 0.58 | 0.58 | 0.56 | 0.58 | 0.65 | 0.71 | 0.73 | 0.71 | 0.66 | 0.61 | 0.58 | 0.54 | 0.52 | 0.67 |
| FBSharesPPost | 0.95 | 0.91 | 0.74 | 0.67 | 0.68 | 0.60 | 0.58 | 0.57 | 0.56 | 0.54 | 0.54 | 0.55 | 0.60 | 0.71 | 0.70 | 0.69 | 0.66 | 0.62 | 0.59 | 0.53 | 0.65 |
| TTShares | 0.81 | 0.56 | 0.50 | 0.48 | 0.53 | 0.58 | 0.55 | 0.53 | 0.57 | 0.54 | 0.52 | 0.53 | 0.50 | 0.50 | 0.51 | 0.51 | 0.51 | 0.51 | 0.51 | 0.50 | 0.54 |
| TTSharesPPost | 0.53 | 0.42 | 0.31 | 0.29 | 0.30 | 0.32 | 0.33 | 0.35 | 0.38 | 0.42 | 0.43 | 0.52 | 0.51 | 0.52 | 0.53 | 0.52 | 0.49 | 0.48 | 0.48 | 0.47 | 0.43 |
| FBShares | 0.59 | 0.27 | 0.21 | 0.24 | 0.25 | 0.27 | 0.28 | 0.29 | 0.32 | 0.35 | 0.33 | 0.32 | 0.42 | 0.57 | 0.59 | 0.63 | 0.65 | 0.65 | 0.64 | 0.60 | 0.42 |
| FBLikes | 0.34 | 0.19 | 0.18 | 0.22 | 0.24 | 0.27 | 0.28 | 0.26 | 0.32 | 0.34 | 0.34 | 0.34 | 0.45 | 0.55 | 0.55 | 0.56 | 0.55 | 0.55 | 0.55 | 0.54 | 0.38 |
| TTLikesPPost | 0.22 | 0.10 | 0.10 | 0.10 | 0.14 | 0.17 | 0.18 | 0.24 | 0.28 | 0.32 | 0.33 | 0.38 | 0.39 | 0.40 | 0.40 | 0.39 | 0.37 | 0.35 | 0.35 | 0.34 | 0.28 |
| TTLikes | 0.20 | 0.01 | 0.05 | 0.02 | 0.10 | 0.19 | 0.25 | 0.31 | 0.36 | 0.37 | 0.35 | 0.35 | 0.34 | 0.34 | 0.34 | 0.34 | 0.34 | 0.33 | 0.33 | 0.32 | 0.26 |
| IGPosts | -0.19 | -0.39 | -0.20 | -0.29 | -0.23 | -0.20 | -0.20 | -0.26 | -0.18 | -0.15 | -0.18 | -0.29 | -0.25 | -0.18 | -0.08 | 0.04 | 0.07 | 0.15 | 0.21 | 0.30 | -0.13 |
| TTPosts | 0.05 | 0.09 | 0.13 | 0.11 | 0.08 | 0.10 | 0.14 | 0.06 | -0.14 | -0.33 | -0.40 | -0.84 | -0.82 | -0.81 | -0.78 | -0.67 | -0.54 | -0.35 | -0.24 | -0.17 | -0.27 |
| FBPosts | -0.87 | -0.89 | -0.88 | -0.88 | -0.85 | -0.87 | -0.85 | -0.91 | -0.95 | -0.96 | -0.96 | -0.95 | -0.94 | -0.94 | -0.93 | -0.90 | -0.88 | -0.86 | -0.83 | -0.80 | -0.89 |

**Table 13**
Pearson correlation results for the SM performance and the Mexican election results.

| Metric - Mexico | Number of Days | | | | | | | | | | | | | | | | | | | | Avg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 14 | 21 | 28 | 30 | 60 | 90 | 120 | 150 | 180 | 210 | 240 | 270 | 300 | |
| TTLikesPPost | 0.97 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 |
| TTSharesPPost | 0.93 | 0.95 | 0.98 | 0.98 | 0.98 | 0.97 | 0.97 | 0.97 | 0.97 | 0.97 | 0.97 | 0.97 | 0.97 | 0.97 | 0.97 | 0.97 | 0.97 | 0.97 | 0.97 | 0.98 | 0.97 |
| FBSharesPPost | 0.98 | 0.94 | 0.89 | 0.87 | 0.90 | 0.65 | 0.70 | 0.88 | 0.84 | 0.90 | 0.92 | 0.95 | 0.95 | 0.95 | 0.95 | 0.96 | 0.96 | 0.96 | 0.96 | 0.96 | 0.90 |
| FBLikesPPost | 0.92 | 0.86 | 0.84 | 0.86 | 0.81 | 0.77 | 0.77 | 0.70 | 0.66 | 0.66 | 0.69 | 0.77 | 0.82 | 0.85 | 0.85 | 0.86 | 0.86 | 0.85 | 0.84 | 0.84 | 0.80 |
| TTLikes | 0.98 | 0.89 | 0.94 | 0.86 | 0.94 | 0.93 | 0.93 | 0.57 | 0.57 | 0.59 | 0.59 | 0.73 | 0.70 | 0.75 | 0.70 | 0.69 | 0.66 | 0.67 | 0.69 | 0.71 | 0.75 |
| TTShares | 0.96 | 0.80 | 0.77 | 0.59 | 0.69 | 0.73 | 0.73 | 0.22 | 0.23 | 0.29 | 0.29 | 0.44 | 0.40 | 0.45 | 0.44 | 0.44 | 0.43 | 0.45 | 0.46 | 0.48 | 0.51 |
| FBShares | 0.93 | 0.43 | -0.21 | -0.35 | -0.15 | -0.42 | -0.32 | -0.21 | -0.22 | 0.21 | 0.36 | 0.86 | 0.94 | 0.95 | 0.96 | 0.97 | 0.98 | 0.98 | 0.97 | 0.97 | 0.43 |
| FBLikes | 0.81 | 0.30 | 0.10 | -0.05 | 0.02 | -0.21 | -0.09 | -0.09 | -0.06 | 0.07 | 0.13 | 0.32 | 0.42 | 0.48 | 0.53 | 0.58 | 0.62 | 0.65 | 0.65 | 0.67 | 0.29 |
| FBCommentsPPost | 0.38 | 0.48 | 0.31 | 0.38 | 0.33 | 0.12 | 0.08 | 0.04 | 0.02 | 0.04 | 0.07 | 0.12 | 0.16 | 0.20 | 0.24 | 0.27 | 0.29 | 0.30 | 0.27 | 0.30 | 0.22 |
| FBComments | 0.20 | 0.01 | -0.44 | -0.55 | -0.54 | -0.79 | -0.85 | -0.39 | -0.32 | -0.29 | -0.27 | -0.13 | -0.09 | -0.07 | -0.04 | -0.03 | 0.01 | 0.03 | 0.02 | 0.05 | -0.23 |
| ~~IGCommentsPPost~~ | ~~-0.32~~ | ~~-0.38~~ | ~~-0.41~~ | ~~-0.38~~ | ~~-0.38~~ | ~~-0.39~~ | ~~-0.40~~ | ~~-0.40~~ | ~~-0.39~~ | ~~-0.40~~ | ~~-0.40~~ | ~~-0.38~~ | ~~-0.39~~ | ~~-0.39~~ | ~~-0.40~~ | ~~-0.42~~ | ~~-0.43~~ | ~~-0.44~~ | ~~-0.45~~ | ~~-0.45~~ | ~~-0.40~~ |
| ~~IGLikesPPost~~ | ~~-0.41~~ | ~~-0.44~~ | ~~-0.45~~ | ~~-0.43~~ | ~~-0.41~~ | ~~-0.42~~ | ~~-0.42~~ | ~~-0.39~~ | ~~-0.38~~ | ~~-0.38~~ | ~~-0.39~~ | ~~-0.39~~ | ~~-0.40~~ | ~~-0.40~~ | ~~-0.40~~ | ~~-0.41~~ | ~~-0.42~~ | ~~-0.42~~ | ~~-0.43~~ | ~~-0.43~~ | ~~-0.41~~ |
| ~~IGComments~~ | ~~-0.47~~ | ~~-0.57~~ | ~~-0.68~~ | ~~-0.64~~ | ~~-0.67~~ | ~~-0.72~~ | ~~-0.73~~ | ~~-0.74~~ | ~~-0.75~~ | ~~-0.75~~ | ~~-0.75~~ | ~~-0.69~~ | ~~-0.62~~ | ~~-0.61~~ | ~~-0.58~~ | ~~-0.58~~ | ~~-0.56~~ | ~~-0.56~~ | ~~-0.56~~ | ~~-0.56~~ | ~~-0.64~~ |
| ~~IGLikes~~ | ~~-0.67~~ | ~~-0.67~~ | ~~-0.73~~ | ~~-0.70~~ | ~~-0.72~~ | ~~-0.74~~ | ~~-0.75~~ | ~~-0.73~~ | ~~-0.74~~ | ~~-0.73~~ | ~~-0.74~~ | ~~-0.70~~ | ~~-0.64~~ | ~~-0.63~~ | ~~-0.59~~ | ~~-0.57~~ | ~~-0.53~~ | ~~-0.53~~ | ~~-0.53~~ | ~~-0.53~~ | ~~-0.66~~ |
| ~~IGPosts~~ | ~~-0.73~~ | ~~-0.74~~ | ~~-0.72~~ | ~~-0.73~~ | ~~-0.71~~ | ~~-0.70~~ | ~~-0.70~~ | ~~-0.70~~ | ~~-0.69~~ | ~~-0.69~~ | ~~-0.69~~ | ~~-0.71~~ | ~~-0.73~~ | ~~-0.74~~ | ~~-0.75~~ | ~~-0.76~~ | ~~-0.77~~ | ~~-0.77~~ | ~~-0.77~~ | ~~-0.77~~ | ~~-0.73~~ |
| FBPosts | -0.76 | -0.99 | -0.95 | -0.94 | -0.91 | -0.87 | -0.80 | -0.90 | -0.89 | -0.87 | -0.86 | -0.76 | -0.53 | -0.51 | -0.46 | -0.52 | -0.51 | -0.52 | -0.53 | -0.58 | -0.73 |
| TTPosts | -0.58 | -0.75 | -0.95 | -0.77 | -0.75 | -0.74 | -0.75 | -0.81 | -0.79 | -0.80 | -0.80 | -0.79 | -0.79 | -0.77 | -0.77 | -0.78 | -0.78 | -0.80 | -0.82 | -0.85 | -0.78 |

To summarize, the absolute number of posts in all countries, irrespective of the platform, presented zero or negative correlations with the electoral results. On the other hand, in all countries, at least 5 of the remaining metrics presented high correlations with the results, in different time windows. The presented data therefore rejects the null hypothesis $H_1$', which validates the alternative hypothesis "$H_1$: It is possible to model the SM performance based on the interactions of users on the official profiles of candidates and find correlations between the SM and the electoral performances of candidates", and answers affirmatively the "RQ1: Is there a correlation between the SM performance of candidates and their electoral performance?"

For RQ2, a further analysis was performed to verify if similar correlations occur among different countries and if the same metrics present the highest correlations in the studied countries. For this, we used two approaches. The first consider the average of the results from all windows, as used to answer RQ1. The second considers the 90-day window, the average time of the campaign in studied countries. Tables 14 and 15 present the five metrics with the highest correlation with the electoral results for each country, respectively, for each analysis.

The results for both approaches are almost the same. In Argentina,

**Table 14**
Higher correlations with electoral results, considering the average of all windows.

| Argentina | | Brazil | |
|---|---|---|---|
| Metric | *r* (averaged) | Metric | *r* (averaged) |
| IGLikesPPost | 0.91 | FBComments | 0.90 |
| TTSharesPPost | 0.86 | IGLikes | 0.87 |
| FBSharesPPost | 0.85 | IGComments | 0.87 |
| TTLikesPPost | 0.85 | IGLikesPPost | 0.86 |
| FBLikesPPost | 0.81 | TTShares | 0.86 |

| **Colombia** | | **Mexico** | |
|---|---|---|---|
| **Metric** | ***r* (averaged)** | **Metric** | ***r* (averaged)** |
| FBComments | 0.83 | TTLikesPPost | 0.98 |
| FBCommentsPPost | 0.83 | TTSharesPPost | 0.97 |
| IGComments | 0.80 | FBSharesPPost | 0.90 |
| IGCommentsPPost | 0.77 | FBLikesPPost | 0.80 |
| IGLikesPPost | 0.75 | TTLikes | 0.75 |

**Table 15**
Higher correlations with electoral results, considering the 90-day window.

| Argentina | | Brazil | |
|---|---|---|---|
| Metric | *r* (90 days) | Metric | *r* (90 days) |
| FBSharesPPost | 0.92 | FBComments | 0.90 |
| IGLikesPPost | 0.91 | IGLikes | 0.88 |
| TTSharesPPost | 0.90 | TTShares | 0.87 |
| TTLikesPPost | 0.88 | IGComments | 0.87 |
| FBLikesPPost | 0.86 | TTLikes | 0.86 |

| **Colombia** | | **Mexico** | |
|---|---|---|---|
| **Metric** | ***r* (90 days)** | **Metric** | ***r* (90 days)** |
| IGComments | 0.94 | TTLikesPPost | 0.98 |
| IGLikes | 0.92 | TTSharesPPost | 0.97 |
| IGLikesPPost | 0.90 | FBSharesPPost | 0.95 |
| FBCommentsPPost | 0.88 | FBShares | 0.94 |
| FBComments | 0.86 | FBLikesPPost | 0.82 |

the metrics with the highest correlation with votes in both approaches were the number of likes and shares per post on all three platforms, the same as Mexico (excluding Instagram). On the other hand, the most correlated metrics in Colombia were related to comments: the absolute number of comments and the number of comments per posts. In Brazil, all platforms presented high correlations, but in opposite to the results of Argentina and Mexico, the absolute number of interactions presented higher correlations than the relative interactions per post.

Considering the average of all windows, only one metric is present for more than 2 countries: Instagram likes per post in Argentina, Brazil, and Colombia, suggesting it as the most important metric. As Instagram metrics are not considered for the analysis of Mexico, it may be considered as the most representative cross-country metric. By considering the smaller window of 90 days, no metric is present for more than 2 countries. This result indicates that even being of the same region and having similar characteristics, each electoral context and use of SM platforms was nation related.

As the main findings regarding patterns among countries, we highlight that: (i) There were zero or negative correlations between the number of posts and electoral results in all countries; and (ii) the number of likes per post on Instagram was the only metric present as one of the most correlated metrics in all countries with available data. Our conclusion is that, although it is useful, these data are not sufficient to reject "$H_2$': It is not possible to find a pattern among different countries between the SM performances of candidates and electoral results".

### 3.3. Discussion

This study has found correlations between SM metrics and election results in all studied countries. This fact is not a novelty, since it has already been stated by many previous studies that claimed success in predicting elections with SM. Tumasjan in his seminal paper (Tumasjan et al., 2010) had already stated that "the mere number of tweets mentioning a political party can be considered a plausible reflection of the vote share". The main difference concerns the definition of SM performance. Most studies have measured performance based on how many people are talking about a candidate (generally on Twitter), but our approach has considered how many people are paying attention to a candidate by interacting with his/her profiles on SM. Thus, this study validates the methodology proposed by (Brito, Meira, & Adeodato, 2021), by extending it to multiple elections and reaching additional conclusions.

This change of perspective addresses many of the challenges observed in previous studies. First, these metrics may be used not only with the most commonly used current SM platforms (*i.e.,* Facebook, Instagram, and Twitter) but even with future platforms which have not yet gained popularity. The only requirement is that the SM being studied works in a wall-based model, instead of direct conversation model (such as Whatsapp or Facebook Messenger).

Moreover, it requires the collection of much less data points, a thousand posts from less than a dozen candidates for each country, instead of millions of posts from the entire population. Furthermore, very few choices have to be made by researchers regarding data collection. Since data are collected from the official profiles of candidates, no keyword for data collection needs to be defined. We proposed that data collection would occur over a long period before elections (10 months), and the use of this data combined with many different window sizes exempts the need of arbitrary decisions regarding how many days of SM data should be collected and used.

The findings of this study may also be used to support work aiming to predict electoral results based on SM data. High correlations of metrics were observed with electoral results, but with variations among countries. As the exact correlations, and consequently the best metrics, may only be obtained after the official election results, we conclude that we should not define, *a priori*, the most suitable SM platform or metrics for predicting elections, since each election may be more related to a specific platform or metric, which may be discovered only after the election. Thus, we reinforce the argument presented in previous research that data from all the most used platforms should be collected, and prediction models must be sufficiently flexible to use all these data and be capable of identifying, throughout the process, the most suitable features. Moreover, these models should consider additional discussions, such as the digital gap and the representativity of using social media as a sample.

This study focused on the analysis of four Latin American countries. They present a similar context, as presented in Section 2, which helps to reduce unintentional bias. Nevertheless, there are also differences, such as population, GDP, and different internet access levels, which may bias the study conclusions. Thus, we focused this analysis not on making general claims, but on specific findings regarding each election in RQ1 and finding similarities and differences in RQ2. Also, the SM scenario is rapidly changing, and the popularization of new platforms such as Tik Tok (Xu, Yan, & Zhang, 2019) may change the scenario for future elections.

Lastly, despite discovering the correlations between interactions and votes, what drives users' interactions on SM profiles of candidates was not the focus of this research. There are some studies in this regard (Heiss, Schmuck, & Matthes, 2019; Xenos, Macafee, & Pole, 2017), but further research is still needed. For example, the widespread of fake news is a current concern (Allcott & Gentzkow, 2017; Spohr, 2017), and to discover the differences between interactions on and propagation of real or fake news may be promising.

## 4. Conclusions

This paper presented a study on the relationship between SM and the electoral performance of candidates running in the presidential elections of the four most populous countries in Latin America: Argentina (2019), Brazil, Colombia, and Mexico (2018). We have used a new approach to find a correlation between the SM performance of candidates and votes received by using derived metrics from the three major social networks: Facebook, Twitter, and Instagram, and considered many different data collection periods: from 1 to 300 days prior to the election date. For this effort, we have collected data from 67,403 posts from candidates within these networks and 449,201,837 interactions on these posts.

Regarding RQ1 (*Is there a correlation between the SM performance of candidates and their electoral performance?*), we have two main findings. First, the absolute number of posts presented zero or negative correlations with the electoral results, irrespective of the country or platform. Second, in all countries, at least 5 of the remaining metrics presented correlations with the results, regardless of the number of days before the elections were considered.

Regarding RQ2 (*Is there a common pattern among countries between the SM performance of candidates and electoral results?*), the most representative pattern was that previously presented regarding the number of posts. Additionally, we found that even being of the same region and having similar characteristics, each electoral context and use of SM platforms was nation related. The number of likes per post on Instagram was the only single metric present among the most correlated metrics in more than two countries.

This study methodology is novel in that it validates a new, recently-defined (Brito, Meira, & Adeodato, 2021) set of SM performance metrics based on data from the three major social networks, which is in contrast to most studies, which focus only on Twitter data. The findings, indicating an Instagram metric as being the most present among countries, reinforces this argument of looking outside Twitter. Also, the absolute amounts normalized by posts were effective performance metrics in all countries. In addition, we used a broader data collection period and, to the best of our knowledge, this is the first study in this area to apply the same methodology for analyzing four different countries, especially in Latin America.

The analyzed elections were held before the COVID-19 pandemic, which led to restrictive measures being adopted worldwide, such as lockdowns and social distancing (Bonaccorsi et al., 2020), accelerating a change in the campaign scenario: from popular rallies and a gathering of supporters to an online campaign. Consequently, it is expected that there will be an increase in the use of SM by candidates in the coming years and in the correlations between the online behavior of citizens and electoral results, and this study is one of the pioneers in this subject.

In terms of future work, we would indicate the use of these metrics and this methodology for testing correlations with data from other elections around the world, especially the elections that occurred during the COVID-19 pandemic, and using different SM platforms. We also point out that further studies using different approaches, such as time series analysis of this data, additional sentiment analysis of posts as performed by (Sandoval-Almazan & Valle-Cruz, 2020), and studies regarding what drives the interactions on candidates' posts, including characteristics such as the rhetoric used, presence of emotions, or even the spreading of fake news, may also be promising. Lastly, we highlight a popular objective of electoral predicting results based on SM data. Thus, we suggest that the same set of metrics may be used as input data for future models; new datasets may be included, such as campaign polls and demographic data; and the use of nonlinear machine learn methods which could be promising.

## CRediT authorship contribution statement

**Kellyton Brito:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft. **Paulo Jorge Leitão Adeodato:** Conceptualization, Methodology, Validation, Writing – review & editing, Supervision.

## Acknowledgments

The authors would like to thank the Fundação de Amparo à Ciência e Tecnologia do Estado de Pernambuco (FACEPE) for the financial support (Grant APQ-1235-1.03/21).

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